Bayesian Logistic Regression for Low Birth Weight Risk Factors

Metropolis-Hastings MCMC from Scratch

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Research Question and Objective

Research question

Which maternal demographic and clinical risk factors are associated with low birth weight, and what is the posterior probability that each factor increases the odds of low birth weight?

Objective

To implement Bayesian logistic regression using a Metropolis-Hastings MCMC sampler built from scratch, estimate posterior distributions of odds ratios, and quantify uncertainty in a way that frequentist confidence intervals cannot.

Primary outcome

- Low birth weight: infant birth weight < 2500g (binary outcome)

Main predictors

- Maternal age, pre-pregnancy weight, race, smoking during pregnancy, history of hypertension, uterine irritability, previous preterm labour

Data Source and Study Design

Data source

- Baystate Medical Center, Springfield, Massachusetts (1986)
- Available as `birthwt` in the R `MASS` package
- Reference: Hosmer, Lemeshow & Sturdivant (2013), *Applied Logistic Regression*

Study design

- Retrospective observational study
- 189 singleton births
- No missing data — complete-case analysis not required

Analytical approach

- Bayesian logistic regression via Metropolis-Hastings MCMC (from scratch)
- Frequentist logistic regression as a benchmark
- Prior sensitivity analysis across three prior scales
- Posterior predictive check for model fit assessment

Cohort Description

Characteristic	Value
Total participants	189
Low birth weight cases (< 2500g)	59 (31.2%)
Normal birth weight	130 (68.8%)
Mean maternal age	23.2 years (median 23)
Mean pre-pregnancy weight	129.8 lbs (median 121)
Missing data	0

Key unadjusted differences between groups

- Smoking prevalence: 50.8% in LBW vs 33.8% in normal birth weight group
- Hypertension: 11.9% vs 3.8%
- Previous preterm labour: 30.5% vs 9.2%
- Uterine irritability: 23.7% vs 10.8%
- Mean maternal weight: 122.1 lbs (LBW) vs 133.3 lbs (normal)

Why Bayesian Logistic Regression?

Bayesian vs. frequentist inference

	Frequentist	**Bayesian**
Parameters	Fixed, unknown	Random variables with distributions
Uncertainty	95% confidence interval	95% credible interval
Interpretation	Interval covers true value 95% of the time across repeated samples	95% posterior probability that the parameter lies in the interval
Prior knowledge	Not incorporated	Explicit prior distribution
Key output	Point estimate + p-value	Full posterior distribution

Practical advantage here

Instead of asking "is the p-value < 0.05?", we can directly answer: "What is the probability that smoking increases the odds of low birth weight?" (Answer: 98.1%)

Bayes' theorem applied to regression

$$\underbrace{p(\beta \mid \mathbf{y})}_{\text{posterior}} \propto \underbrace{p(\mathbf{y} \mid \beta)}_{\text{likelihood}} \times \underbrace{p(\beta)}_{\text{prior}}$$

Likelihood — Bernoulli logistic model

$$p(\mathbf{y} \mid \beta) = \prod_{i=1}^{n} \text{logistic}(\mathbf{x}_i^\top \beta)^{y_i} \left[1 - \text{logistic}(\mathbf{x}_i^\top \beta)\right]^{1-y_i}$$

Prior — weakly informative independent Normal priors

$$\beta_j \sim \mathcal{N}(0,\ 2.5^2) \quad \text{for all } j$$

- Centred at zero (no prior effect); SD $= 2.5$ allows a broad range of odds ratios
- Same default prior used by `rstanarm`; avoids driving extreme estimates in small samples

Metropolis-Hastings MCMC Sampler

Why MCMC?

The posterior $p(\boldsymbol{\beta} \mid \mathbf{y})$ has no closed-form solution for logistic regression. MCMC constructs a Markov chain that converges to the posterior distribution.

Metropolis-Hastings algorithm (implemented from scratch)

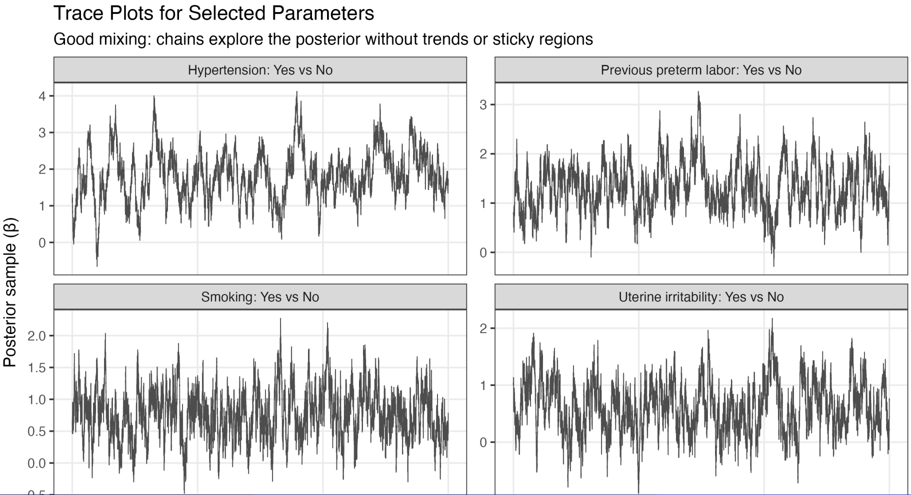
1. Initialise $\boldsymbol{\beta}^{(0)} = \mathbf{0}$
2. At iteration t: propose $\boldsymbol{\beta}^* = \boldsymbol{\beta}^{(t-1)} + \varepsilon$, where $\varepsilon \sim \mathcal{N}(\mathbf{0},\ 0.07^2\mathbf{I})$
3. Compute log acceptance ratio: $\log\alpha = \log p(\boldsymbol{\beta}^* \mid \mathbf{y}) - \log p(\boldsymbol{\beta}^{(t-1)} \mid \mathbf{y})$
4. Accept $\boldsymbol{\beta}^*$ with probability $\min(1, e^{\log\alpha})$; otherwise retain $\boldsymbol{\beta}^{(t-1)}$
5. Repeat for 40,000 iterations; discard first 10,000 as burn-in

Sampler performance

- Post-burn-in samples: 30,000 Acceptance rate: 70.0%
- Trace plots confirm stationarity and good chain mixing

MCMC Convergence: Trace Plots

All four chains show stationary exploration of the posterior with no trends, drifts, or sticky regions — evidence of good convergence and mixing.

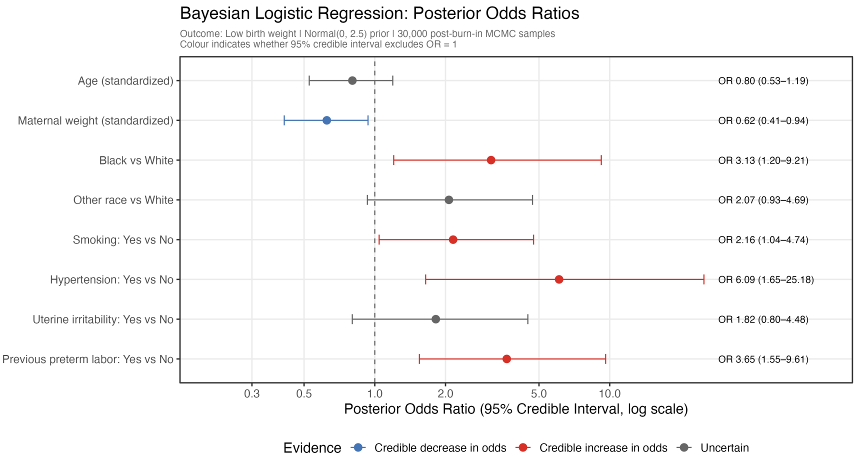


Posterior Summary: All Predictors

Predictor	Posterior OR	95% CrI	P(OR > 1)	Evidence
Age (standardized)	0.80	0.53–1.19	13.1%	Uncertain
Maternal weight (standardized)	0.62	0.41–0.94	1.1%	Credible decrease
Black vs White	3.13	1.20–9.21	99.1%	Credible increase
Other race vs White	2.07	0.93–4.69	96.5%	Uncertain
Smoking: Yes vs No	2.16	1.04–4.74	98.1%	Credible increase
Hypertension: Yes vs No	6.09	1.65–25.18	99.5%	Credible increase
Uterine irritability: Yes vs No	1.82	0.80–4.48	92.0%	Uncertain
Previous preterm labor: Yes vs No	3.65	1.55–9.61	99.9%	Credible increase

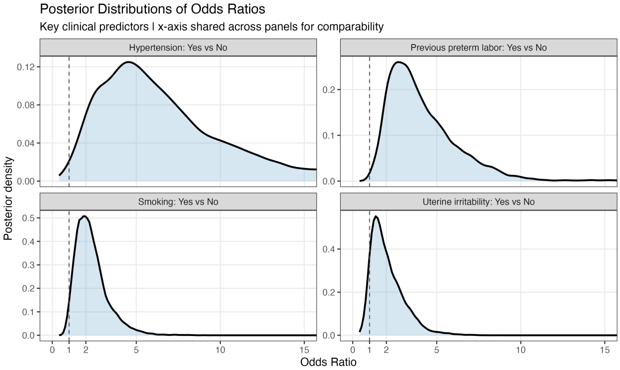
OR = posterior median odds ratio. CrI = 95% credible interval. $P(OR > 1)$ = posterior probability of increased odds. "Credible" indicates the 95% CrI excludes OR = 1.

Forest Plot: Posterior Odds Ratios



Red: 95% CrI entirely above OR = 1 (credible increase). Blue: CrI entirely below 1 (credible decrease). Grey: uncertain (CrI crosses 1). Dashed line: OR = 1.

Posterior Distributions: Key Predictors



Hypertension: wide, right-skewed posterior (n = 12 cases). OR credibly > 1 but magnitude uncertain.

Previous preterm: concentrated; nearly all mass above OR = 1.

Smoking: posterior mass almost entirely above OR = 1.

Uterine irritability: substantial mass below OR = 1; genuine uncertainty.

Shared x-axis (0–15) enables direct cross-predictor comparison.

Clinical Interpretation of Findings

Hypertension

- Posterior OR 6.09 (95% CrI 1.65–25.18); $P(OR > 1) = 99.5\%$ — strongest predictor
- Wide CrI reflects small cell count ($n = 12$); direction robust, magnitude uncertain

Previous preterm labour

- Posterior OR 3.65 (95% CrI 1.55–9.61); $P(OR > 1) = 99.9\%$ — most certain finding

Race (Black vs White)

- Posterior OR 3.13 (95% CrI 1.20–9.21); $P(OR > 1) = 99.1\%$

Smoking during pregnancy

- Posterior OR 2.16 (95% CrI 1.04–4.74); $P(OR > 1) = 98.1\%$

Maternal weight

- Posterior OR 0.62 (95% CrI 0.41–0.94); $P(OR > 1) = 1.1\%$ — only predictor with credible protective effect

Age, other race, uterine irritability

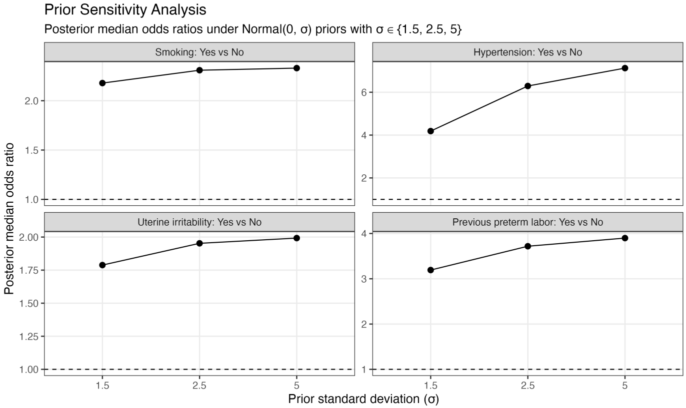
Bayesian vs. Frequentist: Comparison

Predictor	Bayes OR	95% CrI	Freq OR	95% CI
Age (std.)	0.80	0.53–1.19	0.82	0.55–1.20
Maternal weight (std.)	0.62	0.41–0.94	0.63	0.41–0.95
Black vs White	3.13	1.20–9.21	3.36	1.19–9.73
Other race vs White	2.07	0.93–4.69	2.23	0.94–5.50
Smoking: Yes vs No	2.16	1.04–4.74	2.33	1.06–5.29
Hypertension: Yes vs No	6.09	1.65–25.18	6.29	1.64–27.26
Uterine irritability	1.82	0.80–4.48	2.04	0.81–5.06
Previous preterm labor	3.65	1.55–9.61	3.39	1.38–8.60

Key observation

Bayesian and frequentist estimates are consistent in direction and magnitude across all predictors. Bayesian credible intervals are generally slightly narrower due to the regularising effect of the Normal(0, 2.5) prior, particularly for hypertension where the prior pulls the estimate toward zero.

Prior Sensitivity Analysis

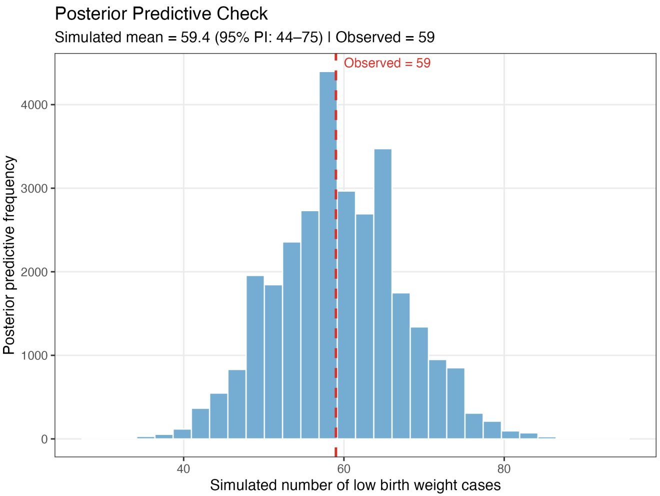


Three prior scales tested

- $\sigma = 1.5$ — tighter prior
- $\sigma = 2.5$ — main analysis
- $\sigma = 5.0$ — diffuse prior

Interpretation

- **Smoking, preterm:** very stable — data dominate the prior
- **Hypertension:** more sensitive, as expected with n = 12 cases
- All four predictors remain above OR = 1 across all prior choices



Method

Simulate new datasets from the posterior. For each draw, generate predicted outcomes and count simulated LBW cases. Compare to the observed count.

Results

- Observed cases: **59**
- Simulated mean: **59.4**
- 95% predictive interval: **44–75**

Interpretation

The observed count falls almost exactly at the simulated mean — no evidence of systematic misfit.

Conclusions

Main findings

- Low birth weight prevalence: 59/189 (31.2%) in this clinical cohort
- Four predictors with credible evidence of increased odds of LBW:
  - Hypertension: OR 6.09 (95% CrI 1.65–25.18); $P(OR > 1) = 99.5\%$
  - Previous preterm labour: OR 3.65 (95% CrI 1.55–9.61); $P(OR > 1) = 99.9\%$
  - Black race vs White: OR 3.13 (95% CrI 1.20–9.21); $P(OR > 1) = 99.1\%$
  - Smoking during pregnancy: OR 2.16 (95% CrI 1.04–4.74); $P(OR > 1) = 98.1\%$
- Higher maternal weight: credibly lower odds — OR 0.62 (95% CrI 0.41–0.94)
- Age, other race, uterine irritability: elevated posterior probabilities (92–97%) but CrIs cross $OR = 1$
- Bayesian and frequentist estimates consistent; findings robust to prior specification
- Posterior predictive check confirms good model fit

Overall conclusion

The MH sampler converged reliably. Bayesian inference provides a richer, more directly interpretable quantification of uncertainty than p-values alone.

Limitations

- Small sample size ($n = 189$); wide credible intervals for several predictors, most notably hypertension ($n = 12$ cases)
- Retrospective observational design from a single centre in 1986; associations cannot be interpreted as causal and may not generalise to other populations or contemporary clinical settings
- The Metropolis-Hastings acceptance rate (70%) is higher than the theoretical optimum for high-dimensional targets (∼23%); while convergence was confirmed visually, a lower proposal variance may improve mixing efficiency
- No formal convergence diagnostics (e.g. Gelman-Rubin $\hat{R}$, effective sample size) were computed; future work could implement multiple chains and automated diagnostics
- Predictor set is limited to variables available in the dataset; important confounders such as prenatal care, socioeconomic status, and nutrition are not captured
- Results should be interpreted as an applied statistical portfolio analysis demonstrating Bayesian methodology

Methods and Tools

Software and packages

- R
- `MASS` — `birthwt` dataset
- `tidyverse` — data wrangling
- `gtsummary` — summary tables
- `broom` — frequentist model output
- `ggplot2` — visualisation

MCMC implementation

- Built from scratch in base R
- No `rstanarm`, `brms`, or `rjags`
- 40,000 iterations, 10,000 burn-in
- Proposal: Normal(0, 0.07)

Statistical methods

- Bayesian logistic regression (MH-MCMC)
- Frequentist logistic regression (benchmark)
- Posterior summary: median OR, 95% CrI, $P(\mathrm{OR} > 1)$
- Prior sensitivity analysis ($\sigma = 1.5, 2.5, 5$)
- Posterior predictive check

MCMC diagnostics

- Trace plots (visual stationarity)
- Acceptance rate monitoring
- Posterior density inspection

Thank You

Questions?

Appendix: Full Posterior Results

Predictor	OR	95% CrI	P(OR>1)	Freq OR	Freq p
Age (standardized)	0.80	0.53–1.19	13.1%	0.82	0.318
Maternal weight (std.)	0.62	0.41–0.94	1.1%	0.63	0.034
Black vs White	3.13	1.20–9.21	99.1%	3.36	0.023
Other race vs White	2.07	0.93–4.69	96.5%	2.23	0.073
Smoking: Yes vs No	2.16	1.04–4.74	98.1%	2.33	0.038
Hypertension: Yes vs No	6.09	1.65–25.18	99.5%	6.29	0.009
Uterine irritability: Yes vs No	1.82	0.80–4.48	92.0%	2.04	0.125
Previous preterm labor: Yes vs No	3.65	1.55–9.61	99.9%	3.39	0.008
MCMC diagnostics					
Post-burn-in samples	30,000				
Acceptance rate	70.0%				
PPC: observed / simulated mean	59 / 59.4 (95% PI: 44–75)				